The Violent Effects of Apex Corruption

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Abstract

We document that *apex* corruption scandals—corrupt acts implicating top-level politicians—cause immediate and persistent increases in violent protests across Latin America. Using a panel of 176 corruption scandals across 23 countries from 2008 to 2018, merged with daily protest counts, we establish three facts: The daily count of violent protests increases by about 59% in the month immediately after the scandal. The effect appears within a week of the scandal, and remains positive even 4 months later. The magnitude of the effect scales sharply with the political rank of the implicated agent: scandals involving sitting presidents show effects larger than those involving Governors or Congress members by an order of magnitude. Our findings identify a high-frequency, high-magnitude channel through which corruption shapes the political environment in democracies.

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1 Introduction

Political violence is among the most consequential outcomes a democracy can experience. Civil unrest disrupts economic activity, displaces investment, and—when met with state violence in return—erodes the social contract that legitimizes the regime. An influencial economic literature has linked political violence to economic shocks (Miguel et al. 2004; McGuirk and Burke 2020), to commodity rents and network structure (König et al. 2017), and in recent work to abrupt withdrawals of foreign assistance (Rohner et al. 2026). The contribution of political shocks to events that change citizens' beliefs about who governs them has received considerably less attention. In this paper, we argue that disclosures of apex corruption are one such shock, and that the behavioral consequences for political violence are large, immediate, and persistent.

Citizens' perceptions of corruption prevalence have risen in the past two decades (Cantoni et al. 2024). Moreover, corruption scandals associated to *apex* politicians have drawn more attention from news outlets in recent years (Rivera et al. 2024); perhaps unsurprisingly due to the expansion of social media, which reduces the cost of diffusing information and enables news outlets to reach broader audiences (Enikolopov et al. 2018; Tufekci 2024). While exposing corruption scandals on news outlets can have positive desired effects, such as political accountability or crackdowns of corruption networks (Ferraz and Finan 2011; Arias et al. 2019, 2022), it can also have negative undesired effects on democratic values (Ajzenman 2021; Rivera et al. 2024) and away from the polling booths. Public exposure of malfeasance by sitting officials changes what citizens believe about who governs them and about the rules under which the government operates. The most consequential corruption-related political moments of the last decade, such as Brazil's Lava Jato[1], were mobilization events, not electoral ones.

We document that apex corruption—corrupt acts implicating top-level politicians—triggers immediate and persistent increases in violent protests across Latin American democracies. Using a panel of 176 corruption scandals across 23 countries from 2008 to 2018, merged with daily protest counts from the Mass Mobilization (MM) project (Clark and Regan 2016), we establish three facts. First, the daily count of violent protests increases by 59% in the month immediately after the scandal, relative to the month immediately preceding the scandal. The count of peaceful protests, however, remains largely unaffected. Second, the effect appears within a week of the scandal break out, and remains positive even 4 months later, relative to the immediate month before the scandal. Third, the magnitude of the effect scales sharply with the political rank of the implicated agent: scandals involving sitting presidents show effects larger than those involving Governors or members of Congress by an order of magnitude. Scandals involving Supreme Court justices, lower judiciary, and

[1] See this video for more information on the scandal.

other non-apex officials leave protest patterns broadly unchanged.

2 Method

2.1 Study Design and Measures

To conduct the empirical analyses we exploit the randomness of corruption scandal breakthroughs in Latinamerican news outlets between 2008 and 2018. In particular, we use a stacked difference-in-differences design in which we compare the evolution of protest occurrences in countries with scandals to those without scandals. Our analysis dataset is created by: (i) Scraping the Twitter accounts from news outlets in Latin America to identify the calendar date in which a scandal in a given country is first reported[2], (ii) create an event-window of $T \in \{30, 120\}$ days before and after the scandal, (iii) merge the country-day count of protests from the MM data to create a 'scandal-protest' dataset for a given country in a given calendar-date window, and lastly, (iv) append all such scandal-protest datasets to build the analysis dataset.

The MM dataset is available for the period 1990–2020, and it includes information on the date, size, and location of any episodes where 50 or more protesters publicly demonstrate against the State. If a protest lasts X days, we code the occurrence of a protest at date $t, t+1, \ldots, t+X-1$. Additionally, if the same protest is associated with more than one date (e.g., because the news source for the protest cannot accurately identify whether it is a continued or a new protest), then we keep the date of the earliest protest. For a protest to be classified as "violent", "[t]he violence could include anything from riotous behavior that destroys property to shooting at the police or military." Our main outcome is the number of protests occurring in a given country-date. We focus on the years 2008–2018, given that our corruption scandals data is available for those years.

We use Ordinary Least Squares to estimate the following equation[3]:

$$Y_{c(s)t} = \beta \, \mathbb{1}\{t \geq \text{Scandal Date}_{c(s)}\} + \alpha_d + \lambda_m + \theta_{cy} + \varepsilon_{c(s)t} \tag{1}$$

where α_d denote day of the week fixed effects, λ_m denote month of the year fixed effects, θ_{cy} denote country-year fixed effects, Scandal Date$_{c(s)}$ is the date in which corruption scandal s in country c occurs, $Y_{c(s)t}$ denotes the outcome of interest in country c at date t, and $\varepsilon_{c(s)t}$ is an idiosyncratic error term.

To trace dynamics we replace $\mathbb{1}\{t \geq \text{Scandal Date}_{c(s)}\}$ in Equation 1 with a saturated set of bin

[2] We keep the date of the first news report to (a) be able to estimate the effect of a scoop, and (b) avoid double-counting a scandal due to follow-up news reports on it.

[3] In the Supplementary Materials we estimate the analogous equation using Poisson Quasimaximum Likelihood to account for the count-nature of the outcome and find even similar effects.

indicators for time to/since scandal. At the narrow window ($T = 30$) we use 6-day bins, which strike a balance between the temporal resolution needed to detect a short-horizon response and the within-bin sample size needed to estimate each bin coefficient precisely. At the wide window ($T = 120$) we use 30-day bins. The reference category in each time-to-event graph is the bin immediately before disclosure (i.e., $w_{s,t} \in [-6, -1]$ at the narrow window, $w_{s,t} \in [-30, -1]$ at the wide window).

The identification assumption we rely on for our estimates to be interpreted as causal is that, conditional on country-year, month-of-year, and day-of-week fixed effects, the within-year date of disclosure of a scandal is as-good-as-randomly assigned in relation to the protest baseline of the country. The country-year fixed effects absorb every source of country-year-level confounding such as pre-existing political instability, electoral calendars, or institutional reforms. The day-of-week and month-of-year fixed effects absorb weekly and seasonal cycles in protest activity.

3 Results

3.1 Apex corruption increases violent protests

Table 1 reports our estimates of the effect of apex corruption on protest occurrences in both the 30- and 120-day windows. Our results show that protests increase after an apex corruption scandal, with increases in violent protests accounting for roughly two-thirds of the overall effect.. In Panel A Column (1) we show that, on average, protests increase by 0.025 protests per day (25% of the mean in the week before the scandal, $P = 0.005$). When we split the protests between violent and peaceful (Columns 2 and 3 of Panel A), we find that apex corruption scandals cause an increase of 0.016 violent protests (59% of the mean in the week before the scandal, $P = 0.001$), whereas we find a statistically insignificant increase of 0.01 peaceful protests following an apex corruption scandal (14% of the mean in the week prior to the scandal, $P = 0.211$). We thus conclude that apex corruption scandals cause people to protest more violently, relative to what the baseline protest would have looked like in the absence of such scandals.

While effects appear immediately within a month of the scandal, we ask whether the effects persist when we consider a longer time horizon. We find that they indeed do. In Panel B of Table 1 we re-estimate the effects of apex corruption scandals on protests using 120-day time windows. We find that protests overall show no statistically significant changes (estimate = 0.007, $P = 0.484$). However, when we analyze violent and peaceful protests separately, we find that violent protests increase by 0.016 (80% of the mean in the month prior to the scandal, $P = 0.008$), while peaceful protests show no statistically significant differences relative to the four months prior to the scandal (if anything, they show a *decrease* of 0.009 protests per day, $P = 0.199$). The slightly negative point estimate for peaceful protests in the long horizon, while imprecise, is consistent with substitution from peaceful to violent mobilization. We cannot, however, identify substitution directly. We next study how these effects evolve over time.

Table 1. Effects of Apex Corruption on Protests

	(1) Any Protests	(2) Violent Protests	(3) Peaceful Protests
Panel A: ±30-Day Window			
Post Scandal	0.025***	0.016***	0.010
	(0.009)	(0.005)	(0.008)
Mean (Pre-Scandal Bin)	0.097	0.027	0.069
Observations	10736	10736	10736
Number of Scandals	176	176	176
R-squared	0.368	0.160	0.473
Panel B: ±120-Day Window			
Post Scandal	0.007	0.016**	-0.009
	(0.010)	(0.006)	(0.007)
Mean (Pre-Scandal Bin)	0.074	0.020	0.053
Observations	42657	42657	42657
Number of Scandals	176	176	176
R-squared	0.231	0.126	0.300
Country × Year FE	✓	✓	✓
SE Cluster	C × Y × DB	C × Y × DB	C × Y × DB

The table reports OLS estimates of β from Equation 1 for three outcome variables: the daily country-level count of any protest events (column 1), violent protests (column 2), and peaceful protests (column 3). An observation is a country-day inside the event window of a scandal; the 176 apex corruption scandals across 23 Latin American countries are stacked. Panel A uses a ±30-day window around each scandal's public-disclosure date; Panel B uses a ±120-day window. All specifications include country-by-year, month-of-year, and day-of-week fixed effects. Standard errors (in parentheses) are clustered at the country × year × day-bin level. "Mean (Pre-Scandal Bin)" reports the average value of the outcome in the bin immediately before the scandal's disclosure—a 6-day bin in Panel A and a 30-day bin in Panel B—computed over the regression sample. The sample period is 2008–2018; Venezuela is excluded. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

3.2 Protest dynamics around apex corruption scandals

In Figure 1 we show estimates of the dynamics of the effects of apex corruption scandals using our modified Equation 1, in which we substitute the indicator for post-scandal dates with a fully saturated set of indicators for 6-day (in the 30-day window) or 30-day (in the 120-day window) bins of dates within the scandal date.

In Figure 1a we show that in the week immediately following apex corruption scandals there is a large increase in the average number of violent protests, which coincides with a sudden stop in the increasing trend of peaceful protests observed in the corresponding time period (Figure 1c). The estimates are thus consistent with the hypothesis that even in periods with protests, apex corruption scandals effectively change the nature of the protests from peaceful to violent, and they do so in as little as 6 days. Importantly, we observe a similar pattern (and in fact, an identical point estimate for the average effect) when we extend the time horizon to 120 days. Violent protests are on a flat trend prior to apex corruption scandals, but they suddenly increase in the months following it (Figure 1b). In contrast, peaceful protest show little to no change in their trend after apex corruption scandals (Figure 1d).

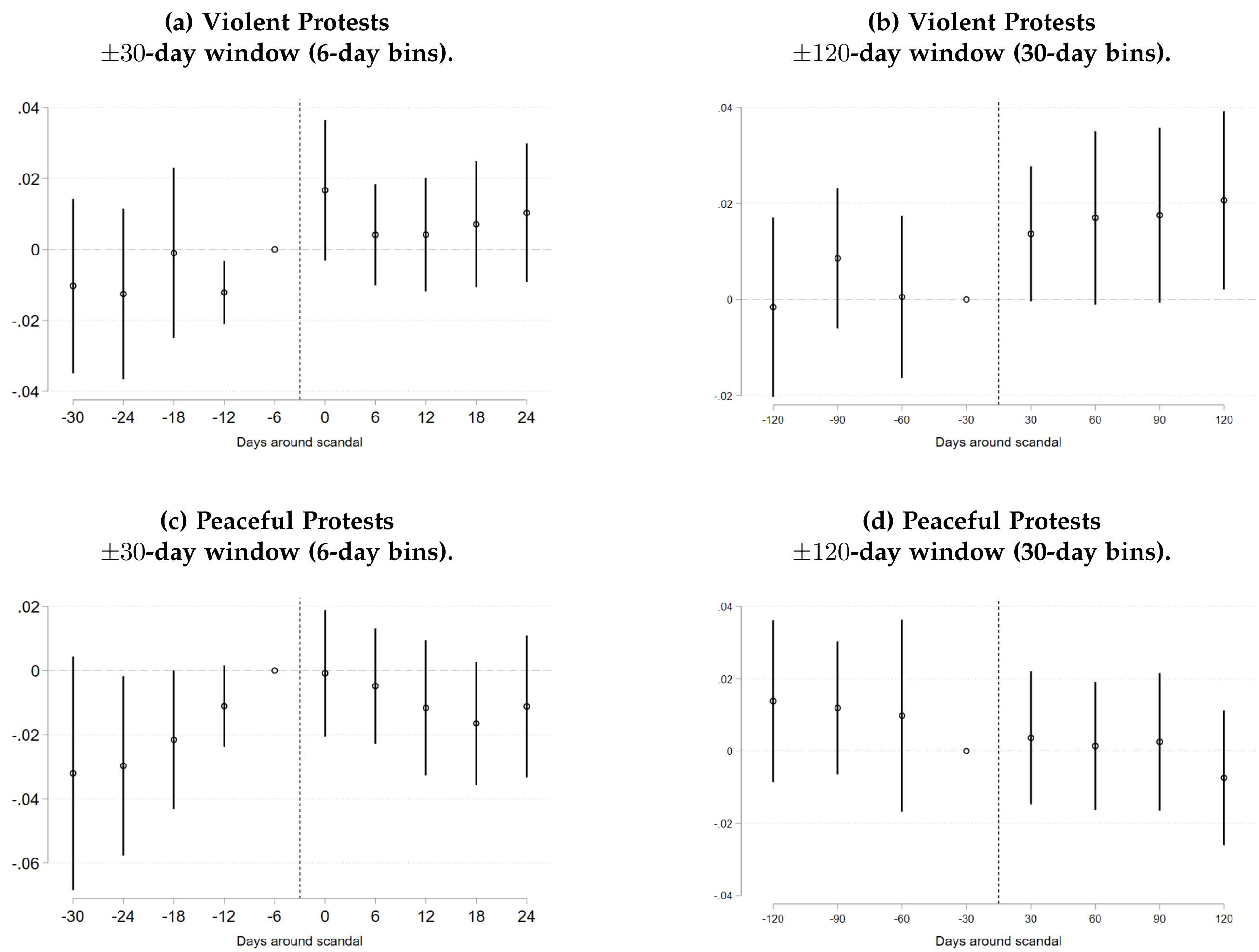
These results together point to two issues. First, citizens' responses to apex corruption scandals materialize fast. In the very first week after the scandal breaks out in the news, protests drastically shift from peaceful to violent. We interpret this rapid shift from peaceful to violent protests as evidence that apex corruption scandals are different in nature from local corruption scandals in that citizens stop trusting democratic institutions, which causes them to seek other forms of expressing their disapproval of the government. Second, apex corruption scandals do not cause one-off protests that fade out easily. Instead, they cause durable protests that persist even four months after the scandal. This speaks to the salience of the stakes and risks associated to tolerating corrupt behavior at the apex of the political spectrum, a topic we further examine next.

3.3 Heterogeneity by political rank

We have thus far showed that apex corruption scandals cause protests to turn violent, and that the effects persist even four months after the scandal breaks out. In this section we show that the effects are heterogeneous even within the apex of the political spectrum. To do this, we classify each of our scandals according to the position of the implicated politician. Then, we estimate Equation 1 for each of our scandals individually on its own country's 30-day window, and plot the distribution of point estimates for each of the categories of implicated politicians. Our sample has 45 scandals in which the implicated politician is a president, and within the remaining 131 scandals, 31 implicate Governors and Members of Congress (we call this group "Other Apex"), and 100 implicate Supreme Court Justices, lower judiciary, and other lower level officials ("Other Non-apex").

Figure 2 characterizes the heterogeneity of the protest response across the political rank of the implicated official. The median effect on any protests for scandals involving Presidents is a striking 0.037 increase in protests, while the corresponding one for scandals implicating Others is a 0.005 *decrease* in protests. Even if we further split the scandals implicating Others into Other Apex and Other Non-apex, we find that the median effect does not differ between the two, which is consistent with the hypothesis that the rank of the implicated official is an important determinant of citizens' reaction to corruption scandals. Even more strikingly, we find that while the distribution of effects on violent

Figure 1. Effects of apex corruption scandals in 6-/30-day bins

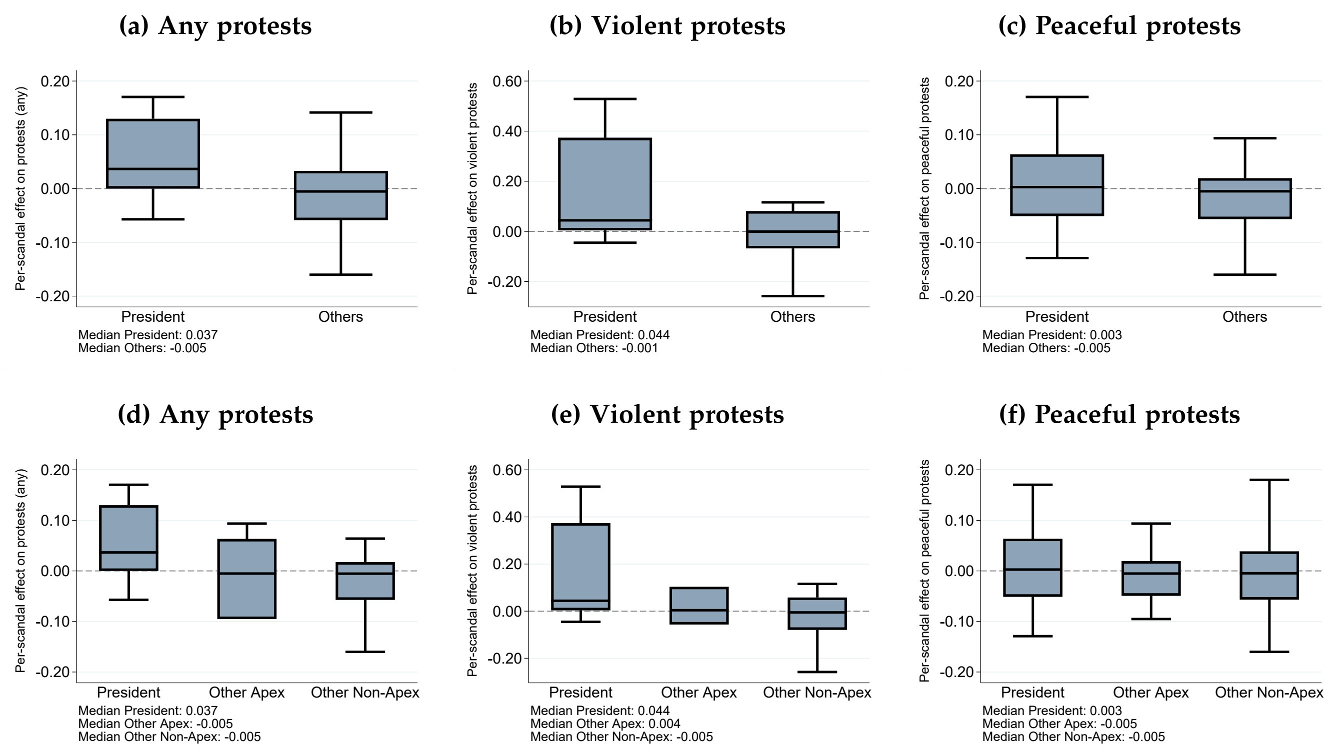


This figure shows the estimates on event-time bin indicators from the modified version of Equation 1 described in Section 2. The omitted reference bin is the one for 1-6 days before the scandal in Panels (a) and (c), and for 1-30 days before the scandal in Panels (b) and (d). Vertical bars are 90% confidence intervals. An observation is a country-scandal-date.

protests is substantially shifted to the right for scandals involving Presidents, the analogous one for peaceful protests is roughly similar across all three categories of implicated officials.

4 Discussion

Figure 2. Distribution of within-scandal effects of apex corruption



For each of the 176 apex corruption scandals, we estimate the OLS post-scandal coefficient of Equation 1 on the ±30-day window of the implicated country alone, retaining the day-of-week and month-of-year fixed effects. Each panel summarizes, by box plot, the cross-scandal distribution of those per-scandal coefficients $\hat{\beta}_s$ for the corresponding outcome. *Top row*: scandals naming a sitting President (45 scandals) against the pooled non-Presidential category (127 scandals; Governors, members of Congress, judicial actors, and lower officials). *Bottom row*: the three-way apex partition—President (45 scandals), Other Apex (27 scandals; Governors and members of Congress), and Other Non-Apex (100 scandals; Supreme Court justices, lower judiciary, and other lower officials). Within each row, the three columns correspond to the three outcomes: any protests (left), violent protests (center), and peaceful protests (right). The note below each box plot reports the within-category median of $\hat{\beta}_s$. Outliers are hidden for legibility. The sample period is 2008–2018; Venezuela is excluded.

References

Ajzenman, Nicolás, "The Power of Example: Corruption Spurs Corruption," *American Economic Journal: Applied Economics*, April 2021, *13* (2), 230–57.

Arias, Eric, Horacio Larreguy, John Marshall, and Pablo Querubín, "Priors Rule: When Do Malfeasance Revelations Help Or Hurt Incumbent Parties?," *Journal of the European Economic Association*, 05 2022, *20* (4), 1433–1477.

_ , Pablo Balán, Horacio Larreguy, John Marshall, and Pablo Querubín, "Information Provision, Voter Coordination, and Electoral Accountability: Evidence from Mexican Social Networks," *American Political Science Review*, 2019, *113* (2), 475–498.

Cantoni, Davide, Andrew Kao, David Y. Yang, and Noam Yuchtman, "Protests," *Annual Review of Economics*, 2024, *16*, 519–543.

Clark, David and Patrick Regan, "Mass Mobilization Protest Data," 2016.

Enikolopov, Ruben, Maria Petrova, and Konstantin Sonin, "Social Media and Corruption," *American Economic Journal: Applied Economics*, January 2018, *10* (1), 150–174.

Ferraz, Claudio and Frederico Finan, "Electoral Accountability and Corruption: Evidence from the Audits of Local Governments," *American Economic Review*, June 2011, *101* (4), 1274–1311.

König, Michael D., Dominic Rohner, Mathias Thoenig, and Fabrizio Zilibotti, "Networks in conflict: theory and evidence from the Great War of Africa," *Econometrica*, 2017, *85* (4), 1093–1132.

McGuirk, Eoin and Marshall Burke, "The economic origins of conflict in Africa," *Journal of Political Economy*, 2020, *128* (10), 3940–3997.

Miguel, Edward, Shanker Satyanath, and Ernest Sergenti, "Economic shocks and civil conflict: an instrumental variables approach," *Journal of Political Economy*, 2004, *112* (4), 725–753.

Rivera, Eduardo, Enrique Seira, and Saumitra Jha, "Apex Corruption Erodes Democratic Values," *Working Paper*, 2024.

Rohner, D., U. Sunde, O. Vanden Eynde, A. L. Wright, and J.-R. Zeng, "Aiding peace or conflict? The impact of USAID cuts on violence," *Science*, 2026, *392* (6799), eaed6802.

Tufekci, Zeynep, *Twitter and tear gas*, New Haven, CT: Yale University Press, 2024.